SET C

Register No	R	A	2	5	1	2	0	4	4	0	1	5	1	2	4

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSC555J – MACHINE LEARNING ALGORITHMS
CYCLE TEST – I
Theory

Program Offered: M. Tech
Max. Marks: 40

Year / Semester: I/ I
Duration: 1.5 Hrs
Date of Exam: 27/09/25

Part A (2 x 20 = 40 marks)

Answer any two questions

1 (i) Explain how linear regression works using the mathematical expression for a sample dataset? Also, what are the different performance evaluation metrics, and how do they work? **15**

(ii) Compare Supervised, Unsupervised, and Semi-Supervised Learning. Provide examples of when each type of learning is most suitable. **5**

2 (i). Explain in detail about logistic regression classifier, with suitable example **10**

(ii). What is loss function, explain how Gradient descent helps to Minimize the loss **10**

3 (i) Describe the K-fold Cross-Validation process. How does it work, and why is it more robust than a simple train-test split? Provide a scenario where K-fold CV would be useful. **10**

(ii) What are the different performance metrics of a Classification model? Explain the concept of confusion matrix and compute the below measure with definition **10**

N=195	No	Yes
No	65	43
Yes	32	55

a) Accuracy

b) Precision

c) Recall

d) F1-score

2
3 6
4 8
5 10
6 12
7 14
8 16